Embassies and Democracy

Steven Pappas and Gregory J. Matthews

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#Summary

Model details

Reduced model

$$Y_{kt} \sim \text{Poisson}(\lambda_{kt})$$

$$\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1kt} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$$

$$\gamma_{0k} \sim \text{N}(0, \sigma_k^2)$$

$$\gamma_{0t} \sim \text{N}(0, \sigma_t^2)$$

where Y_{kt} is the number of embassies for country k in year t and X_{1kt} is one of the democracy scores for country k in year t. We also considered models where the democracy score as a predictor was lagged by 1, 2, or 3 years (i.e. for a single lag: $\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1k(t-1)} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$ where $X_{1k(t-1)}$ is the democracy score in the prior year.)

FIUX THIDS: As shown in Table 1, in these reduced models none of the democracy scores or their lags display significance at an $\alpha = 0.05$ confidence level.

Full model

The full models (results in Table 2) follow a similar setup as the reduced models, but they include additional explanatory variables for the country's GNI, population, region, colonization status after World War II, ideology, and whether the year is during the Cold War. We applied log transformations to GNI and population, since they displayed a right-skewed distribution. We also kept the reduced models' random effects for country and year and followed a similar setup for lagged democracy scores, going back as far as three years. The model equation is given below, with k and t again representing country and year, repsectively.

$$Y_{kt} \sim \text{Poisson}(\lambda_{kt})$$

$$\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1kt} + ... + \beta_7 X_{7kt} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$$

$$\gamma_{0k} \sim \text{N}(0, \sigma_k^2)$$

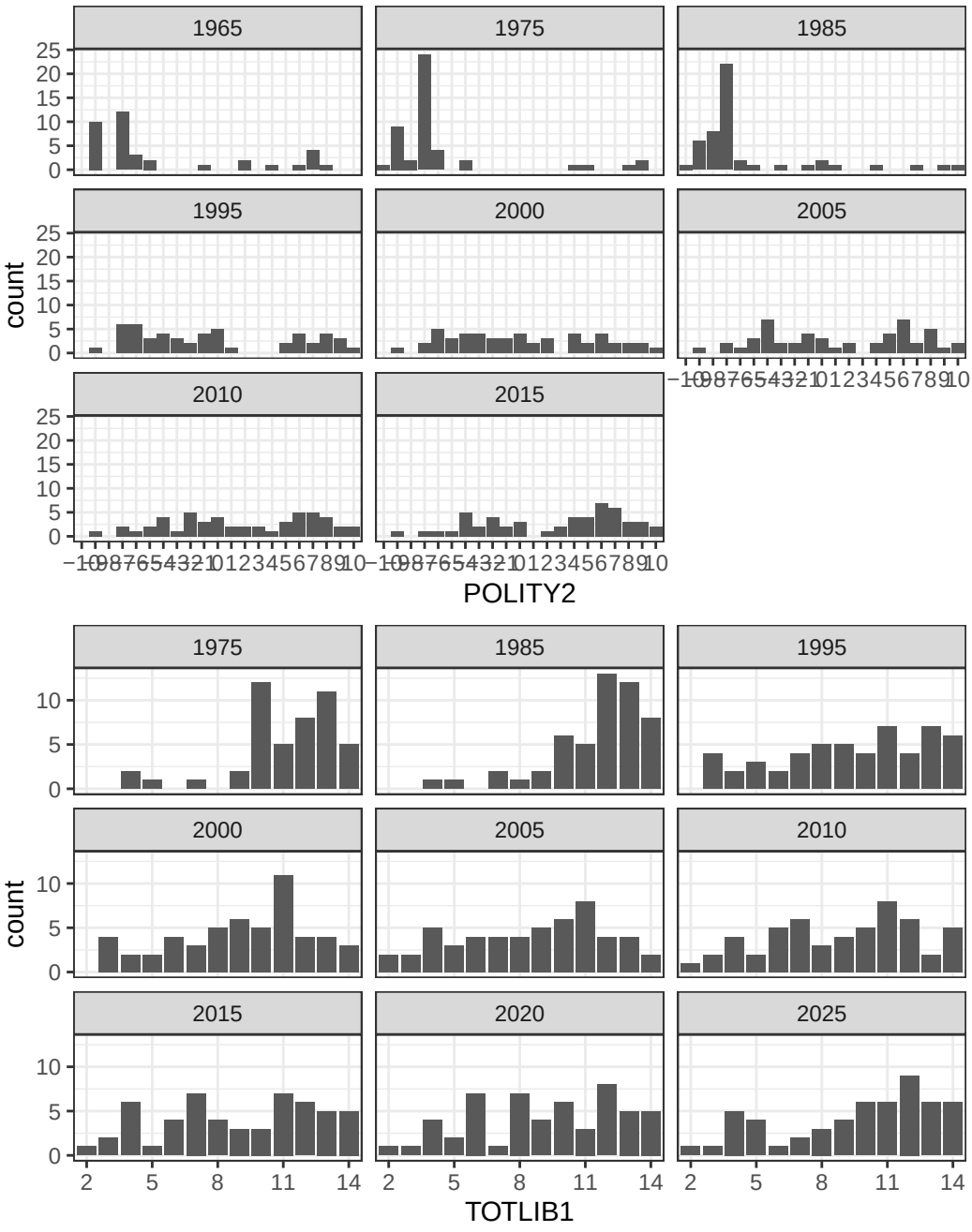
$$\gamma_{0t} \sim \text{N}(0, \sigma_t^2)$$

where again Y_{kt} is the number of embassies for country k in year t and X_{1kt} is one of the democracy scores for country k in year t. The additional X's represent country's GNI, population, region, colonization status after World War II, ideology, and whether the year is during the Cold War. We also considered models where the democracy score as a predictor was lagged by 1, 2, or 3 years (i.e. for a single lag: $\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1k(t-1)} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$ where $X_{1k(t-1)}$ is the democracy score in the prior year.)

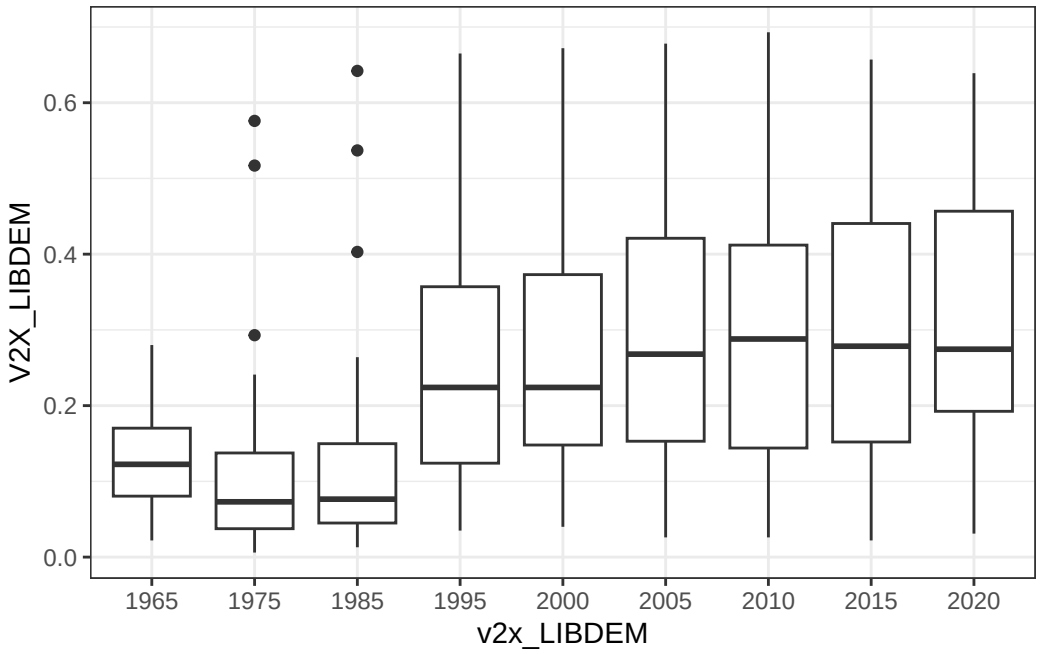
Similar to the reduced models, in the full models none of the democracy scores or their lags display significance at an $\alpha = 0.05$ confidence level. Additionally, none reached a p-value under 0.2. Full results are shown in Table 2.

Exploratory Data Analysis

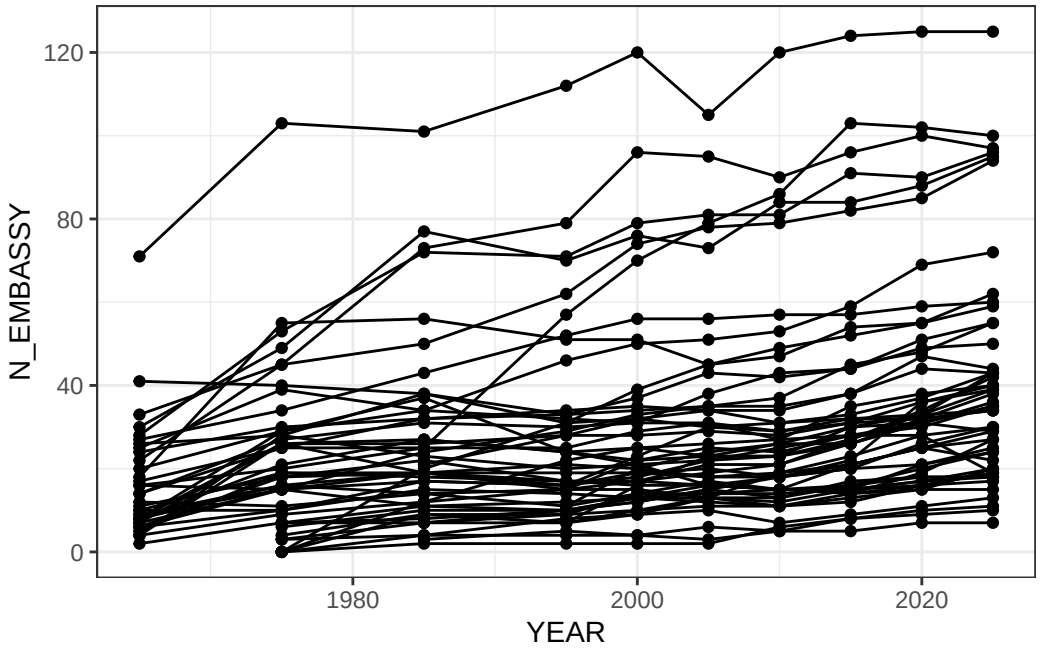
Below are the distributions of the three democracy variables.


1965
1975
1985
1995
2000
2005
2010
2015
count
POLITY2

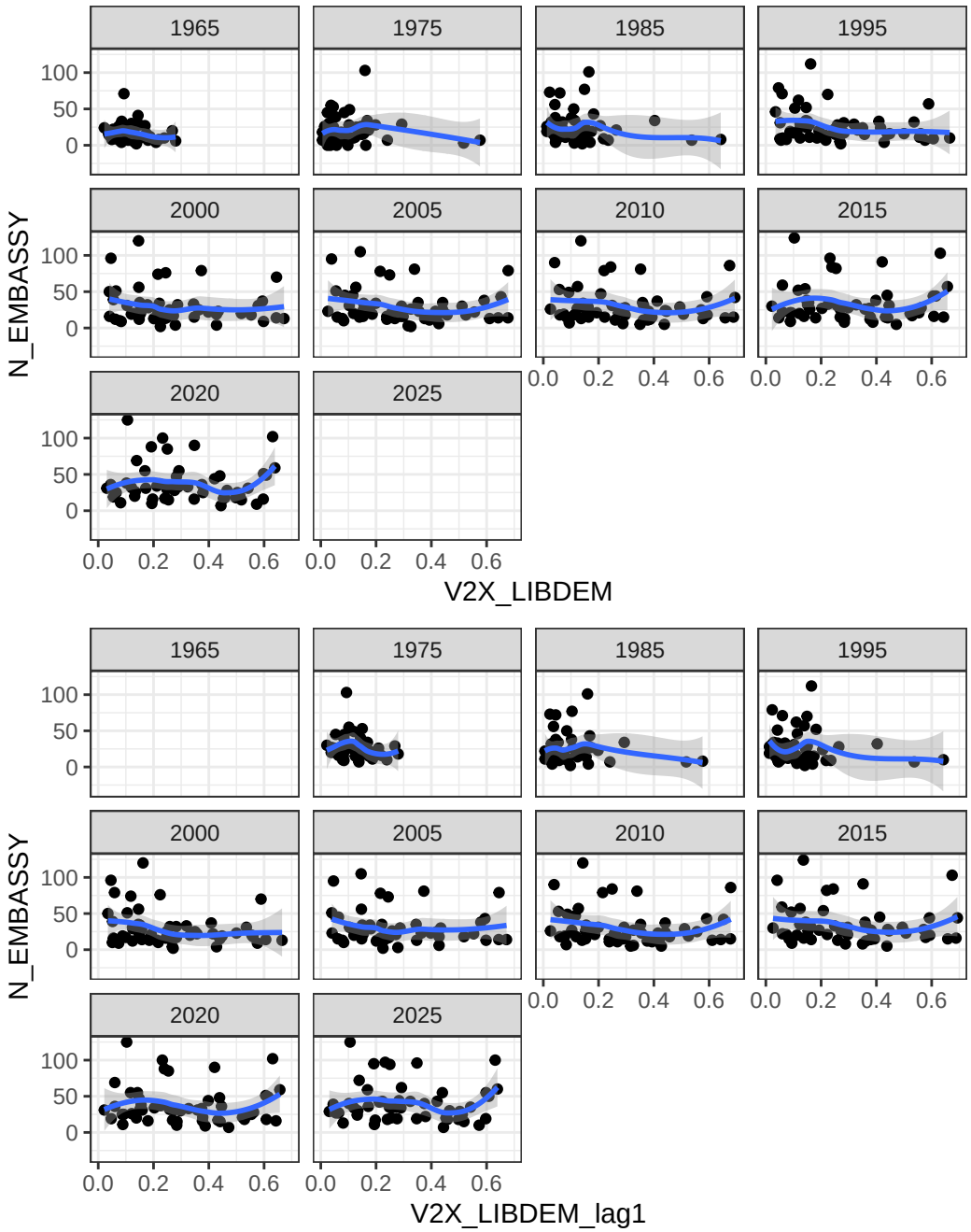
1975
1985
1995
2000
2005
2010
2015
2020
2025
count
TOTLIB1



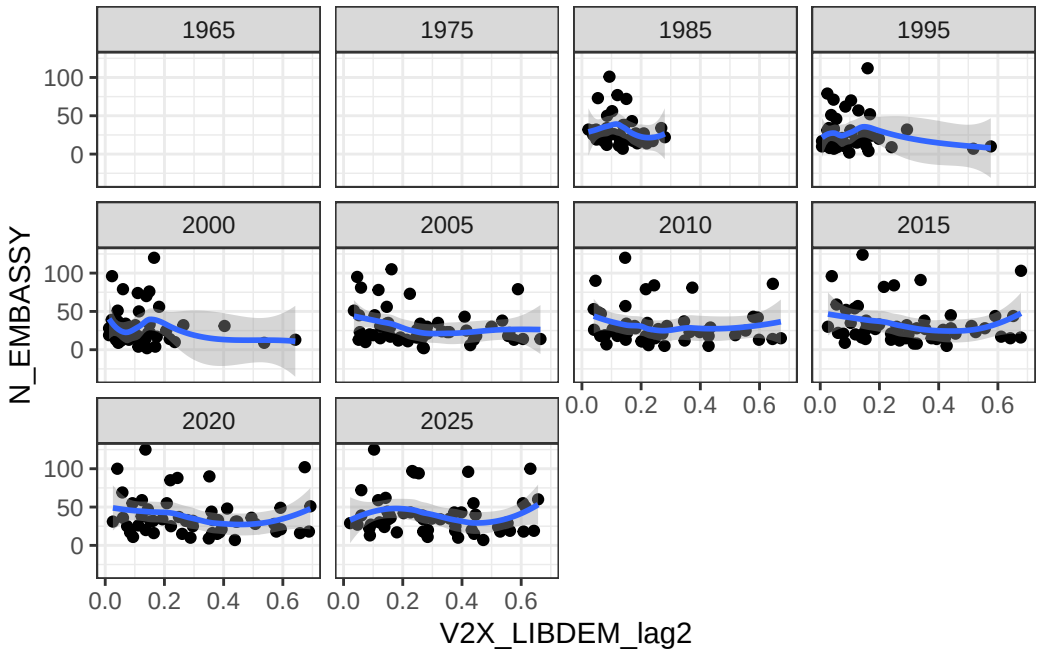
The plot below is the number of embassies over time with each connected line representing a single country.

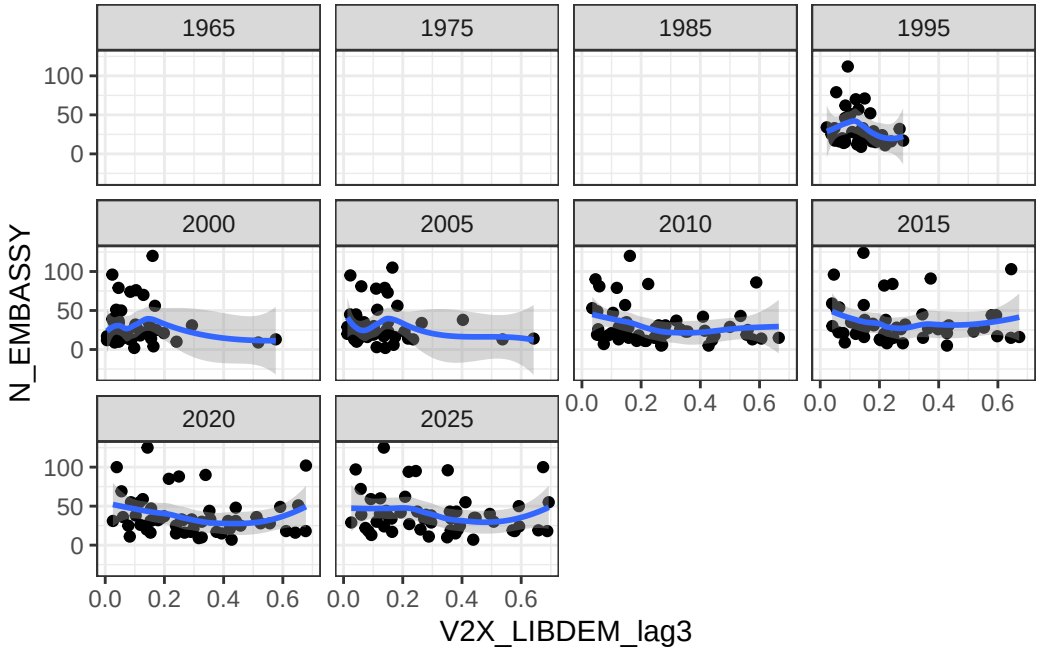


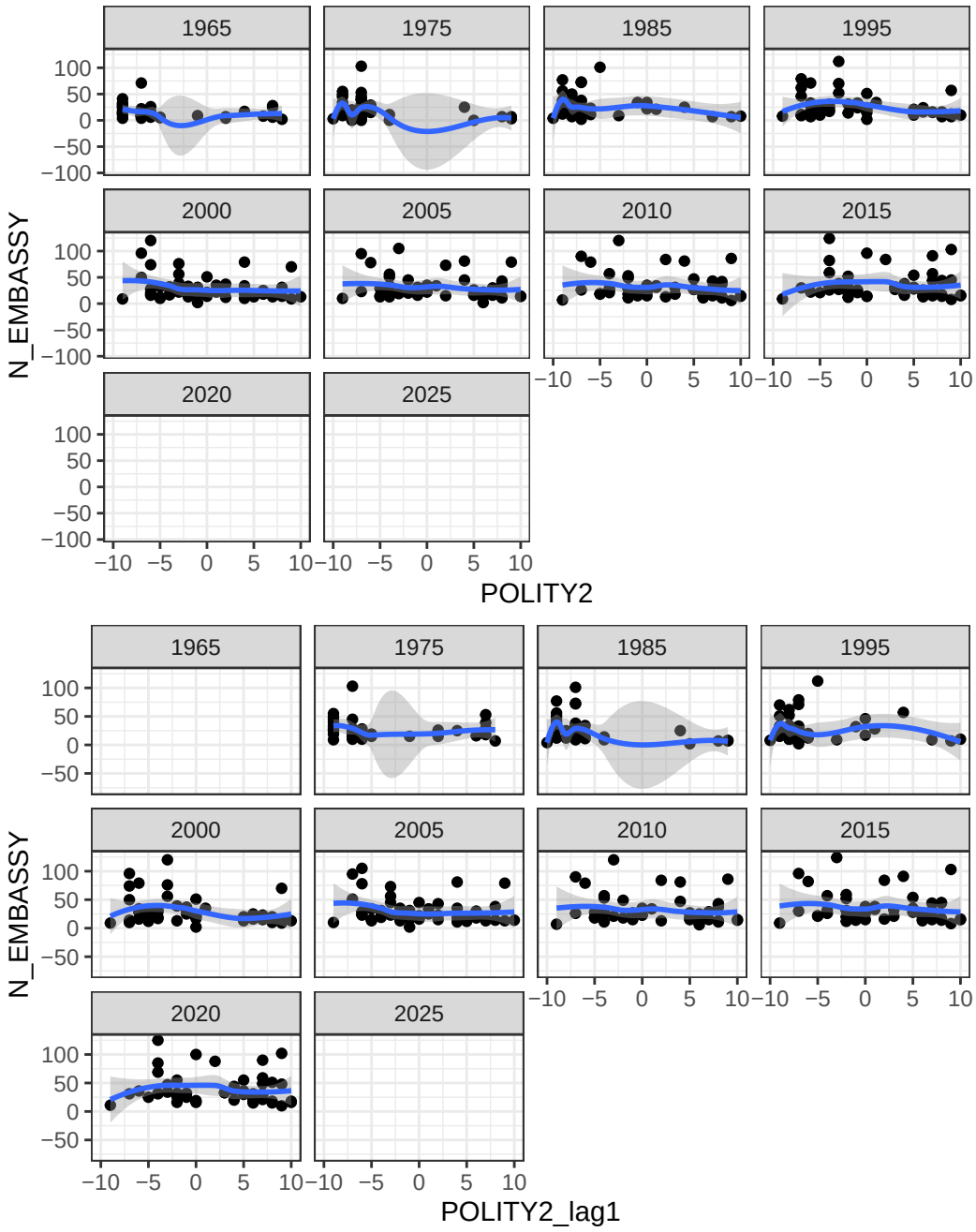
Next, scatter plots of the number of the democracy variables versus the embassies are plotted and each plot includes a trend line of the relationship.

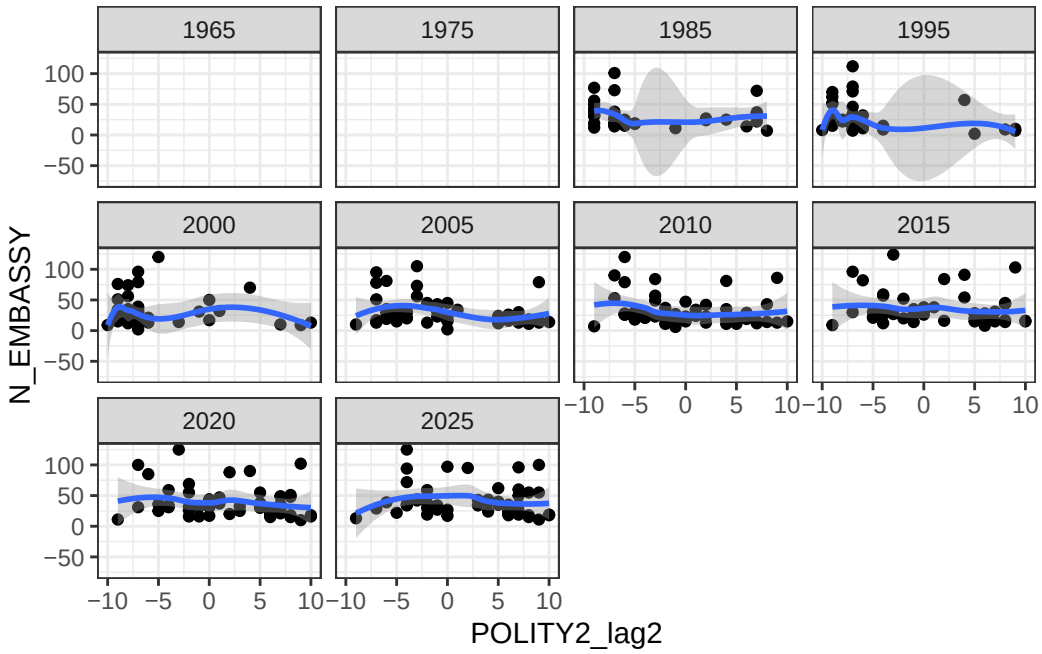

1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
V2X_LIBDEM
0
50
100
0.0
0.2
0.4
0.6

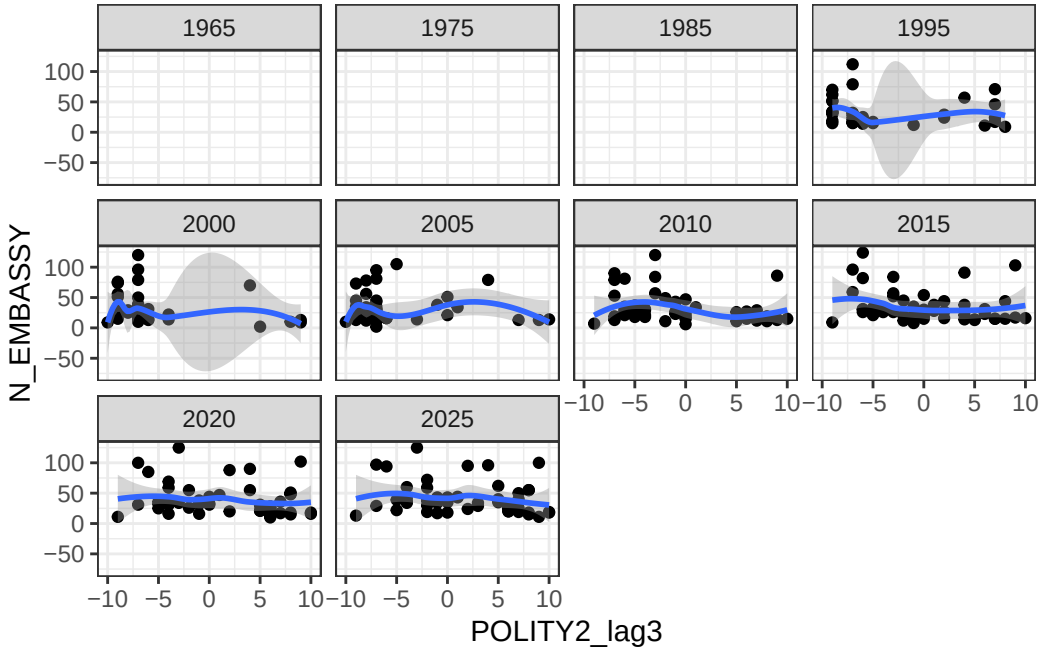
1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
V2X_LIBDEM_lag1
0
50
100
0.0
0.2
0.4
0.6

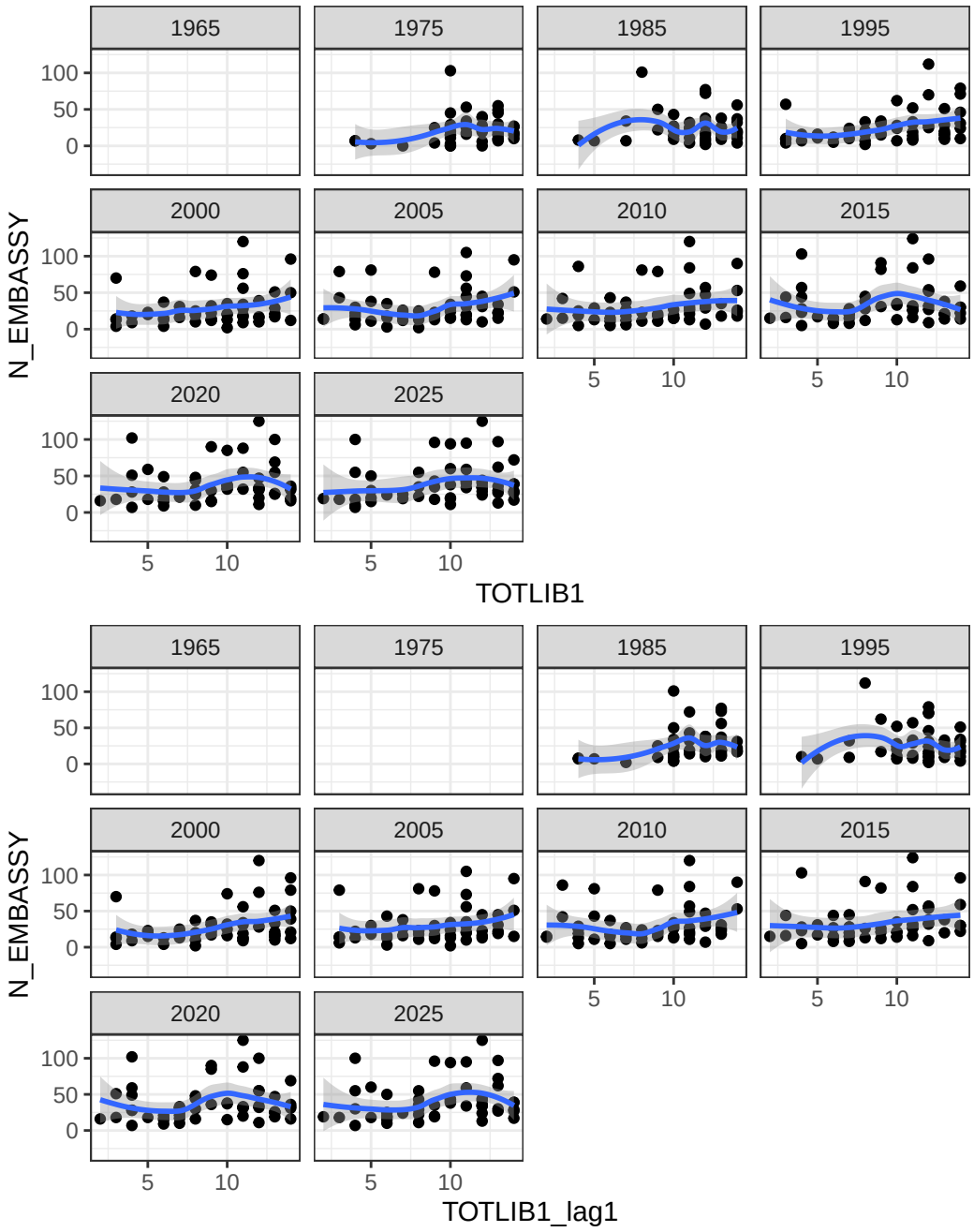

1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
100
50
0
0.0
0.2
0.4
0.6
V2X_LIBDEM_lag2

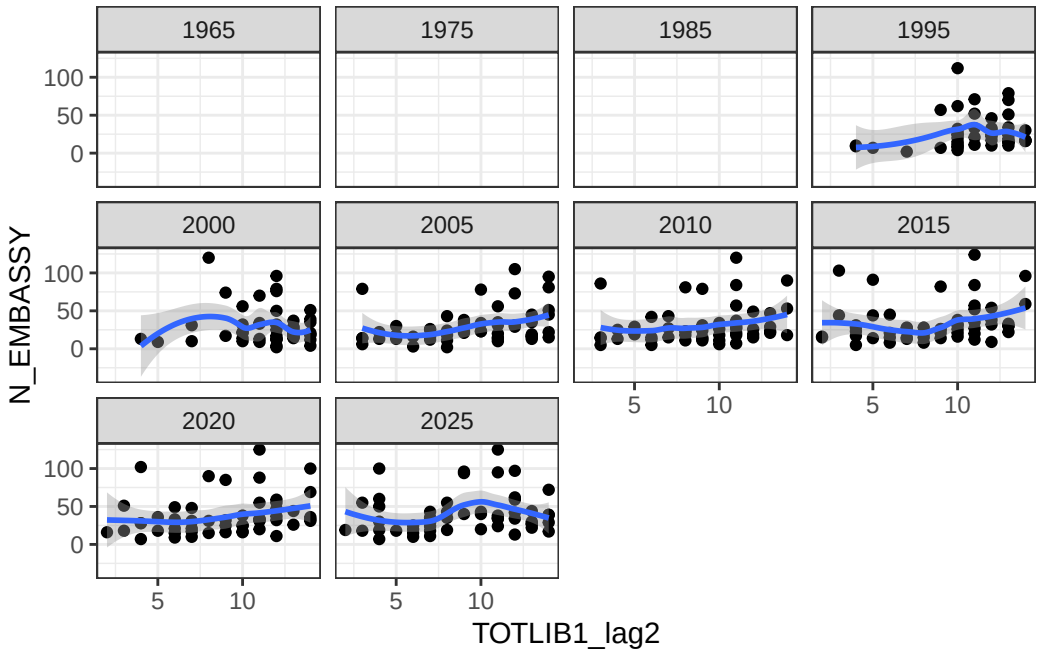

1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
100
50
0
0.0
0.2
0.4
0.6
V2X_LIBDEM_lag3

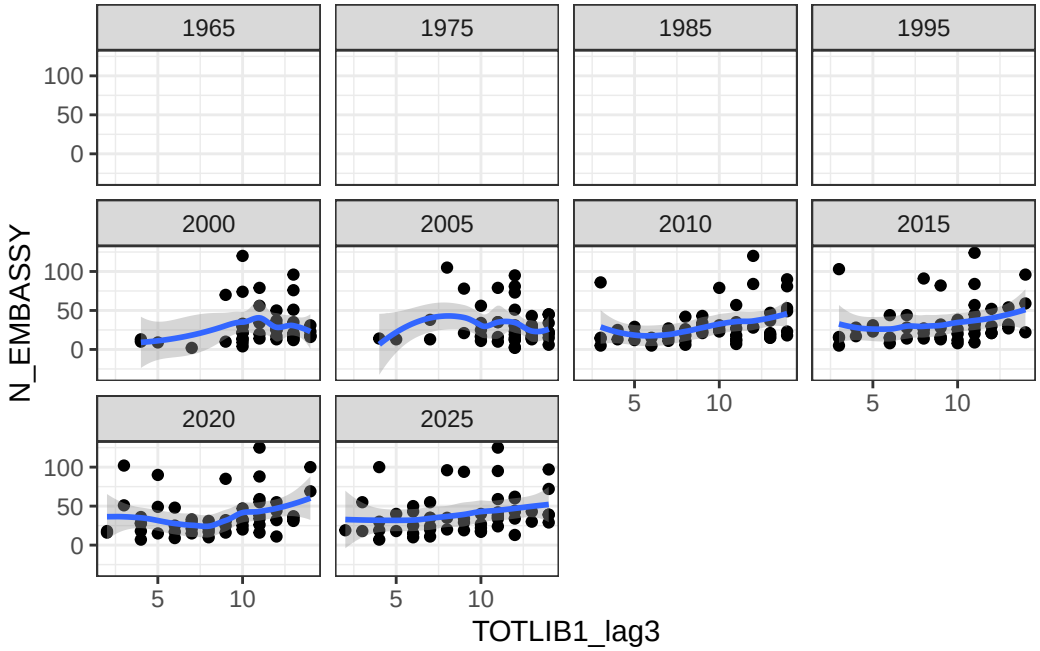

1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
POLITY2
POLITY2_lag1


1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
POLITY2_lag2


1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
POLITY2_lag3


1965
1975
1985
1995
2000
2005
2010
2015
2020
2025
N_EMBASSY
TOTLIB1
TOTLIB1_lag1
0
50
100
5
10






```
masterset %>% filter(!is.na(POLITY2)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, POLITY2, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, POLITY2)$p.val
)
```

```
# A tibble: 8 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1965 -0.222  0.186
2  1975 -0.305  0.0370
3  1985 -0.199  0.170
4  1995 -0.284  0.0437
5  2000 -0.253  0.0735
6  2005 -0.209  0.141
7  2010 -0.198  0.164
8  2015 -0.0826 0.560
```

```
masterset %>% filter(!is.na(POLITY2_lag1)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, POLITY2_lag1, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, POLITY2_lag1)$p.val
)
```

```
# A tibble: 8 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1975 -0.121  0.476
2  1985 -0.341  0.0190
3  1995 -0.0992 0.498
4  2000 -0.273  0.0530
5  2005 -0.236  0.0954
6  2010 -0.188  0.186
7  2015 -0.173  0.224
8  2020 -0.120  0.395
```

```
masterset %>% filter(!is.na(POLITY2_lag2)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, POLITY2_lag2, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, POLITY2_lag2)$p.val
)
```

```
# A tibble: 8 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1985 -0.134  0.429
2  1995 -0.250  0.0903
3  2000 -0.0738 0.614
4  2005 -0.278  0.0485
```

```
5  2010 -0.229  0.106
6  2015 -0.167  0.240
7  2020 -0.203  0.153
8  2025 -0.122  0.391
```

```
masterset %>% filter(!is.na(POLITY2_lag3)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, POLITY2_lag3, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, POLITY2_lag3)$p.val
)
```

```
# A tibble: 7 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1995 -0.0907 0.593
2  2000 -0.213  0.150
3  2005 -0.0367 0.802
4  2010 -0.258  0.0679
5  2015 -0.214  0.132
6  2020 -0.194  0.172
7  2025 -0.208  0.143
```

```
masterset %>% filter(!is.na(V2X_LIBDEM)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, V2X_LIBDEM, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, V2X_LIBDEM)$p.val
)
```

```
# A tibble: 9 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1965 -0.176  0.306
2  1975 -0.0921 0.538
3  1985 -0.115  0.425
4  1995 -0.262  0.0579
5  2000 -0.129  0.357
6  2005 -0.146  0.295
7  2010 -0.128  0.362
8  2015 -0.0257 0.854
9  2020 -0.0542 0.697
```

```r
masterset %>% filter(!is.na(V2X_LIBDEM_lag1)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, V2X_LIBDEM_lag1, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, V2X_LIBDEM_lag1)$p.val
)
```

```
# A tibble: 9 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1975 -0.197  0.248
2  1985 -0.127  0.395
3  1995 -0.100  0.488
4  2000 -0.240  0.0829
5  2005 -0.0940 0.503
6  2010 -0.133  0.343
7  2015 -0.115  0.413
8  2020 -0.0569 0.683
9  2025 -0.0716 0.607
```

```r
masterset %>% filter(!is.na(V2X_LIBDEM_lag2)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, V2X_LIBDEM_lag2, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, V2X_LIBDEM_lag2)$p.val
)
```

```
# A tibble: 8 x 3
   YEAR     cor  pval
  <dbl>   <dbl> <dbl>
1  1985 -0.159  0.353
2  1995 -0.111  0.458
3  2000 -0.0946 0.513
4  2005 -0.219  0.115
5  2010 -0.0823 0.558
6  2015 -0.120  0.390
7  2020 -0.142  0.311
8  2025 -0.0783 0.574
```

```r
masterset %>% filter(!is.na(V2X_LIBDEM_lag3)) %>% group_by(YEAR) %>% summarize(
  cor = cor(N_EMBASSY, V2X_LIBDEM_lag3, use = "pairwise.complete.obs") ,
  pval = cor.test(N_EMBASSY, V2X_LIBDEM_lag3)$p.val
)
```

```
# A tibble: 7 x 3
   YEAR     cor  pval
  <dbl>   <dbl> <dbl>
1  1995 -0.191  0.263
2  2000 -0.110  0.462
3  2005 -0.0819 0.572
4  2010 -0.198  0.154
5  2015 -0.0722 0.608
6  2020 -0.144  0.304
7  2025 -0.148  0.290
```

```
masterset %>% filter(YEAR > 1965 &
                       YEAR < 2024 &
                       !is.na(TOTLIB1)) %>% group_by(YEAR) %>% summarize(
                         cor = cor(N_EMBASSY, TOTLIB1, use = "pairwise.complete.obs") ,
                         pval = cor.test(N_EMBASSY, TOTLIB1)$p.val
                       )
```

```
# A tibble: 8 x 3
   YEAR     cor    pval
  <dbl>   <dbl>   <dbl>
1  1975  0.218  0.141
2  1985 -0.0334 0.816
3  1995  0.372  0.00612
4  2000  0.242  0.0812
5  2005  0.208  0.135
6  2010  0.217  0.118
7  2015  0.104  0.455
8  2020  0.180  0.193
```

```
masterset %>% filter(YEAR > 1965 &
                       YEAR < 2024 &
                       !is.na(TOTLIB1_lag1)) %>% group_by(YEAR) %>% summarize(
                         cor = cor(N_EMBASSY, TOTLIB1_lag1, use = "pairwise.complete.obs") ,
                         pval = cor.test(N_EMBASSY, TOTLIB1_lag1)$p.val
                       )
```

```
# A tibble: 7 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1985  0.250  0.0896
```

```
2  1995 -0.0768 0.592
3  2000  0.341  0.0123
4  2005  0.217  0.119
5  2010  0.191  0.171
6  2015  0.211  0.128
7  2020  0.137  0.323
```

```r
masterset %>% filter(YEAR > 1965 &
                       YEAR < 2024 &
                       !is.na(TOTLIB1_lag2)) %>% group_by(YEAR) %>% summarize(
                         cor = cor(N_EMBASSY, TOTLIB1_lag2, use = "pairwise.complete.obs") ,
                         pval = cor.test(N_EMBASSY, TOTLIB1_lag2)$p.val
                       )
```

```
# A tibble: 6 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  1995  0.158  0.290
2  2000 -0.0803 0.575
3  2005  0.321  0.0192
4  2010  0.208  0.136
5  2015  0.184  0.188
6  2020  0.244  0.0779
```

```r
masterset %>% filter(YEAR > 1965 &
                       YEAR < 2024 &
                       !is.na(TOTLIB1_lag3)) %>% group_by(YEAR) %>% summarize(
                         cor = cor(N_EMBASSY, TOTLIB1_lag3, use = "pairwise.complete.obs") ,
                         pval = cor.test(N_EMBASSY, TOTLIB1_lag3)$p.val
                       )
```

```
# A tibble: 5 x 3
   YEAR     cor   pval
  <dbl>   <dbl>  <dbl>
1  2000  0.131  0.380
2  2005 -0.0874 0.542
3  2010  0.309  0.0243
4  2015  0.211  0.129
5  2020  0.218  0.117
```

Table 1: Internal Analysis: Reduced Model Summary for V2X_LIBDEM

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	3.014***	0.142	3.145***	0.111	3.180***	0.107	3.220***	0.106
V2X_LIBDEM	0.017	0.099						
V2X_LIBDEM_lag1			0.149	0.094				
V2X_LIBDEM_lag2					0.201*	0.100		
V2X_LIBDEM_lag3							0.201+	0.110
SD (Intercept YEAR)	0.303		0.173		0.138		0.123	
SD (Intercept CCODE)	0.706		0.673		0.671		0.669	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 2: Internal Analysis: Reduced Model Summary for POLITY2

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	3.004***	0.148	3.184***	0.108	3.279***	0.102	3.317***	0.101
POLITY2	−0.014***	0.003						
POLITY2_lag1			−0.006*	0.003				
POLITY2_lag2					−0.003	0.003		
POLITY2_lag3							−0.001	0.003
SD (Intercept YEAR)	0.328		0.175		0.160		0.139	
SD (Intercept CCODE)	0.661		0.627		0.609		0.611	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: Internal Analysis: Reduced Model Summary for TOTLIB1

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
(Intercept)	3.205***	0.125	3.248***	0.119	3.309***	0.118	3.327***	0.118
TOTLIB1	−0.004	0.005						
TOTLIB1_lag1			−0.003	0.005				
TOTLIB1_lag2					−0.006	0.005		
TOTLIB1_lag3							−0.003	0.006
SD (Intercept YEAR)	0.193		0.155		0.138		0.121	
SD (Intercept CCODE)	0.691		0.677		0.674		0.667	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Below are the summaries for the full models for ODAG from external countries. As with the reduced model summaries, the $\widehat{\beta}$ coefficients are starred by significance as denoted in each table, with their respective standard error below.

Table 4: Internal Analysis: Full Model Summary for V2X_LIBDEM

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.385	0.270	0.627*	0.269	0.665*	0.280	0.773**	0.295
V2X_LIBDEM	−0.048	0.098						
V2X_LIBDEM_lag1			−0.072	0.105				
V2X_LIBDEM_lag2					0.003	0.112		
V2X_LIBDEM_lag3							0.059	0.114
logPOP	0.435***	0.066	0.448***	0.069	0.512***	0.075	0.570***	0.081
logGNI	0.296***	0.051	0.224***	0.052	0.165**	0.058	0.143*	0.061
logTRDIMtot	0.102*	0.041	0.087+	0.046	0.054	0.049	−0.008	0.052
logTRDEXtot	0.038	0.038	0.063	0.042	0.084+	0.045	0.085+	0.047
COLD_WAR	0.101	0.070	0.156**	0.056	0.120+	0.065		
CW_Socialist	0.106+	0.061	−0.003	0.078	0.075	0.095		
CW_Marxist	−0.165+	0.089	−0.202*	0.089	−0.048	0.125		
COL_British	0.021	0.088	0.040	0.090	0.045	0.094	0.049	0.101
COL_Portuguese	0.062	0.126	0.050	0.128	0.072	0.134	0.080	0.143
COL_Spanish	0.144	0.244	0.123	0.251	0.178	0.263	0.264	0.284
COL_Italian	0.590***	0.177	0.614***	0.182	0.674***	0.192	0.668**	0.213
COL_Belgian	0.025	0.237	0.054	0.243	0.023	0.255	0.001	0.275
COL_German	−0.162	0.111	−0.162	0.114	−0.156	0.119	−0.156	0.129
COL_US	0.321	0.230	0.159	0.237	0.053	0.251	−0.013	0.272
COL_None	0.068	0.240	0.065	0.246	0.025	0.257	−0.025	0.276
COL_Unique	0.443+	0.258	0.400	0.263	0.367	0.272	0.329	0.289
REG_E_Africa	−0.752***	0.131	−0.755***	0.135	−0.778***	0.142	−0.819***	0.153
REG_S_Africa	−0.609***	0.139	−0.629***	0.143	−0.667***	0.151	−0.724***	0.161
REG_C_Africa	−0.327*	0.144	−0.372*	0.147	−0.440**	0.155	−0.536**	0.166
REG_W_Africa	−0.388**	0.124	−0.434***	0.127	−0.487***	0.135	−0.561***	0.144
SD (Intercept YEAR)	0.074		0.047		0.042		0.036	
SD (Intercept CCODE)	0.203		0.208		0.218		0.235	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 5: Internal Analysis: Full Model Summary for POLITY2

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.295	0.296	0.705*	0.287	0.673*	0.296	0.806*	0.317
POLITY2	−0.007*	0.003						
POLITY2_lag1			−0.008**	0.003				
POLITY2_lag2					−0.007*	0.003		
POLITY2_lag3							−0.004	0.003
logPOP	0.409***	0.075	0.389***	0.075	0.441***	0.079	0.508***	0.087
logGNI	0.347***	0.055	0.256***	0.052	0.218***	0.058	0.182**	0.062
logTRDIMtot	0.105*	0.044	0.084+	0.045	0.064	0.048	0.007	0.052
logTRDEXtot	0.019	0.042	0.061	0.041	0.076+	0.044	0.078	0.048
COLD_WAR	0.076	0.073	0.128*	0.054	0.123*	0.061		
CW_Socialist	0.080	0.063	−0.022	0.078	0.038	0.096		
CW_Marxist	−0.190*	0.090	−0.223*	0.090	−0.089	0.126		
COL_British	0.049	0.088	0.072	0.091	0.075	0.094	0.075	0.102
COL_Portuguese	0.144	0.132	0.131	0.136	0.149	0.140	0.152	0.151
COL_Spanish	−0.029	0.246	0.036	0.250	0.086	0.257	0.177	0.279
COL_Italian	0.573***	0.174	0.566**	0.180	0.624***	0.187	0.635**	0.209
COL_Belgian	0.054	0.231	0.089	0.240	0.047	0.247	0.005	0.270
COL_German	−0.183+	0.110	−0.169	0.112	−0.160	0.116	−0.158	0.126
COL_US	0.415+	0.225	0.179	0.234	0.088	0.243	0.016	0.266
COL_None	0.123	0.236	0.090	0.243	0.046	0.249	−0.005	0.270
COL_Unique	0.506+	0.285	0.410	0.260	0.371	0.265	0.340	0.283
REG_E_Africa	−0.744***	0.130	−0.715***	0.134	−0.716***	0.139	−0.780***	0.151
REG_S_Africa	−0.610***	0.139	−0.633***	0.144	−0.643***	0.149	−0.714***	0.161
REG_C_Africa	−0.263+	0.143	−0.336*	0.147	−0.375*	0.152	−0.478**	0.165
REG_W_Africa	−0.350**	0.122	−0.418***	0.126	−0.440***	0.131	−0.530***	0.141
SD (Intercept YEAR)	0.074		0.043		0.037		0.038	
SD (Intercept CCODE)	0.195		0.205		0.210		0.228	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 6: Internal Analysis: Full Model Summary for TOTLIB1

	Unlagged		One-Year Lag		Two-Year Lag		Three-Year Lag	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Intercept	0.548*	0.274	0.609*	0.279	0.701*	0.299	0.569+	0.300
TOTLIB1	0.000	0.006						
TOTLIB1_lag1			0.008	0.006				
TOTLIB1_lag2					−0.001	0.006		
TOTLIB1_lag3							0.003	0.006
logPOP	0.454***	0.069	0.486***	0.073	0.573***	0.080	0.486***	0.079
logGNI	0.227***	0.052	0.181***	0.055	0.151*	0.060	0.184*	0.090
logTRDIMtot	0.109*	0.043	0.060	0.048	−0.005	0.051	0.053	0.078
logTRDEXtot	0.047	0.040	0.083+	0.044	0.091+	0.047	0.104*	0.051
COLD_WAR	0.162**	0.058	0.111+	0.062				
CW_Socialist	−0.007	0.077	0.061	0.093				
CW_Marxist	−0.190*	0.091	−0.157	0.113				
COL_British	0.029	0.091	0.052	0.093	0.050	0.100	0.050	0.094
COL_Portuguese	0.047	0.129	0.067	0.132	0.081	0.141	0.096	0.133
COL_Spanish	0.136	0.252	0.147	0.257	0.236	0.278	0.172	0.262
COL_Italian	0.592**	0.183	0.602**	0.187	0.663**	0.202	0.682***	0.198
COL_Belgian	0.052	0.245	0.023	0.250	−0.023	0.271	−0.079	0.255
COL_German	−0.161	0.115	−0.146	0.117	−0.155	0.127	−0.151	0.119
COL_US	0.175	0.240	0.056	0.247	−0.008	0.269	−0.015	0.253
COL_None	0.091	0.248	0.049	0.252	−0.023	0.272	−0.033	0.253
COL_Unique	0.429	0.267	0.353	0.270	0.349	0.286	0.343	0.268
REG_E_Africa	−0.791***	0.135	−0.786***	0.138	−0.817***	0.149	−0.721***	0.144
REG_S_Africa	−0.636***	0.144	−0.660***	0.147	−0.712***	0.158	−0.660***	0.149
REG_C_Africa	−0.362*	0.149	−0.443**	0.151	−0.511**	0.163	−0.434**	0.159
REG_W_Africa	−0.438***	0.128	−0.472***	0.130	−0.540***	0.141	−0.441**	0.135
SD (Intercept YEAR)	0.051		0.039		0.035		0.051	
SD (Intercept CCODE)	0.211		0.214		0.232		0.211	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001